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Spectral Properties of Phenylbenzopyrylium Salts in Solutions as a Result of Their Nucleophilic Solvation

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A dependence of spectral properties of 2-phenylbenzopyrylium (PBP) salts in solutions on the solvent nucleophilicity is presented. A simple solvate complex model for quantum chemical estimation of the nucleophilic solvation influence on band positions in absorption and fluorescence PBP salt spectra is proposed. Quantum yield changes of the PBP salts in solvents of different nucleophilicity are explained as due to the influence of nucleophile solvation on the efficiency of the intersystem crossing in PBP cations. Two ways of this influence are proposed: the *direct* one, due to energy changes in S_1 and T_i levels, and the *indirect* one, due to an increase or decrease of spin-orbital interaction efficiency, caused by changes of the structural rigidity of solvated PBP cations.